

PROJECT REPORT

Animated 3D Farm Scene with Lighting and Transformations

Graphics Lab Project

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Technology	OpenGL

1. Project Topic

The project is a small animated 3D farm environment developed using OpenGL. It focuses on fundamental computer graphics concepts, particularly **3D transformations, object animation, and basic lighting**. The scene will demonstrate how translation, rotation, and scaling can be applied to different objects while lighting is used to create a more natural outdoor environment.

2. Project Objects

- **Farmhouse** – the main structure of the farm scene.
- **Tractor** – moves through the farm using translation and demonstrates animation.
- **Windmill** – rotating blades demonstrate continuous rotational transformation.
- **Trees and Crops** – repeated/scaled objects used to build the farm environment.
- **Ground and Fence** – basic environmental geometry defining the farm area.
- **Sun / Sky** – provides the visual setting and directional illumination for the scene.

3. Core Graphics Concepts

- **Translation:** used for moving the tractor and clouds.
- **Rotation:** used for the windmill blades and other animated elements.
- **Scaling:** used to create trees, crops, and objects of different sizes.
- **Lighting:** basic ambient, diffuse, and specular lighting will be demonstrated.
- **Animation:** selected objects will move continuously or respond to user input.

4. Extra Features

- **Day/Night Mode:** switch between different lighting conditions.
- **Interactive Camera:** allow the user to rotate or move the camera around the farm.
- **Animated Clouds:** clouds move slowly across the sky to add motion to the scene.
- **Dynamic Windmill:** blade rotation speed can be controlled.
- **Simple User Controls:** keyboard controls for animation and scene interaction.

5. Expected Outcome

The final application will present a compact, visually understandable 3D farm scene in which the major objects are constructed from basic OpenGL geometry. The project will demonstrate the relationship between transformations, animation, and lighting in an interactive graphics environment.

6. Scope

The project is intentionally limited to a small environment so that the implementation remains manageable for a graphics laboratory project while still demonstrating the required computer graphics concepts clearly.